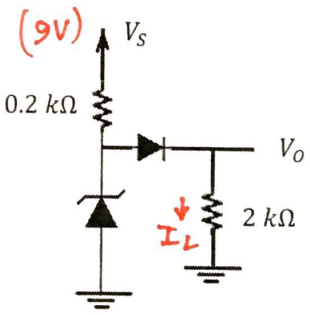
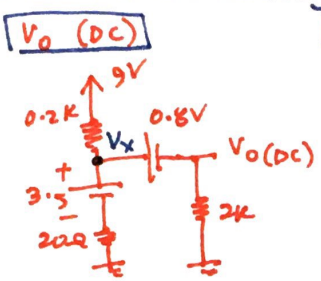
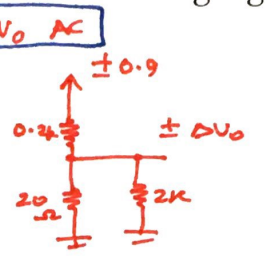
1883 — 75 minutes (40+30+30 points)

Name:	ID:	$V_S = 9$ V	$V_{DD} = 9$ V	Mark: /100

1) The circuit below uses a Zener diode with $V_{ZO} = 3.5\,V$, $r_z = 20\,\Omega$ and $I_{z_min} = 1.0\,mA$. The regular diode is modeled with $V_{DO} = 0.8\,V$ and $r_D = 0\,\Omega$ when forward-biased. (a) If the source V_S has a ±10% variation, calculate the percentage variation in the output voltage V_O. (b) Find the minimum load resistance that can be connected without losing regulation. Show all steps.







$V_{O(DC)} = V_X - 0.8$
$= 3.17\,V$

$\pm\Delta V_O = \pm(0.9)\dfrac{2k//20}{0.2\,k + 2k//20}$
$= \pm\,0.081\,V$

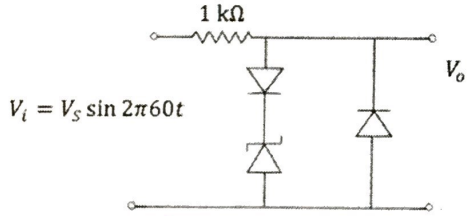
(a)	(b)

$\%\,\dfrac{\Delta V_O}{V_{O(DC)}} = 2.56\,\%$

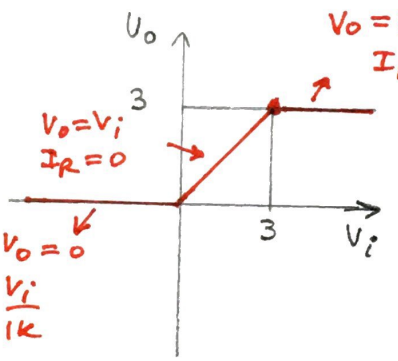
$I_L = \dfrac{0.9(9) - V_{O\,DC}}{0.2k} - 1$

$R_{L\,min} = V_O / I_L$

2) For the diode circuit below (a) Draw the transfer function (V_O vs. V_i) of the circuit. Show all labels on the curve. (b) Find the maximum and minimum current through the resistor. Use the ideal diode model. The Zener diode has $V_{ZO} = 3\,V$ and $r_z = 0\,\Omega$. Show all steps and calculations.



$(V_S = 9\,V)$



$V_O = 3\,V$
$I_R = \dfrac{V_i - 3}{1k}$

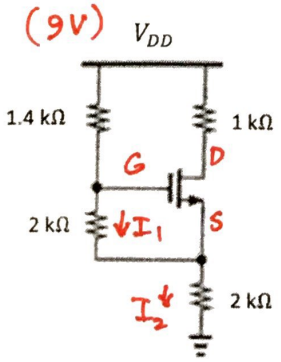
$I_R = \dfrac{V_i}{1k}$

I_{R_max}	I_{R_min}

$I_{R\,min} = -9\,mA$

$I_{R\,max} = 6\,mA$

3) The MOSFET in the circuit below has $V_t = 1$ V, k=1 mA/V² and λ=0. Given I_D=1.5 mA, analyze the circuit. Show all steps and calculations.



Assume saturation

$1.5m = 1m\,(V_{GS} - 1)^2$

$\Rightarrow V_{GS} = 2.22\,V$

$I_1 = V_{GS}/2k = 1.11\,mA$

$I_2 = I_1 + I_D = 2.61\,mA$

$V_S = I_2\,(2k) = 5.22\,V$

$V_G = 2.22 + 5.22 = 7.44\,V$

$V_D = 9 - 1.5m(1k) = 7.5\,V$

} saturation ✓

V_D	V_G	V_S